Visus: Text-Guided Face Occlusion Segmentation with CLIP Attention and Cross-Modal Fusion

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https://github.com/SreenityaThatikunta/Visus

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Abstract

Given a face image and a text prompt such as "glasses" or "face mask", Visus returns a pixel mask of the region the named object covers. It uses CLIP [1]'s own attention as a visual prior: the CLS-to-patch attention of the last vision block marks roughly where the object is. A lightweight segmentation head then refines this prior, using the CLIP text embedding of the prompt through cross-modal attention. Because no public dataset has pixel labels for these occlusions, we build one: 29,668 CelebA-HQ faces with landmark-aligned face masks and procedurally drawn glasses, where the labels are exact because the occlusions are composited by our own code. On a held-out test split, which uses mask textures never seen in training, the model reaches an occlusion IoU of 0.952 for glasses and 0.979 for face masks. When asked for the occluder that is not in the image ("glasses" on a mask-only face), it correctly returns an empty mask 99.7% of the time. Ablations show that fine-tuning CLIP, feeding the image to the head and using the full text embedding each matter. This report also describes the first version of the project (v1) and the bugs we found in it. Its training, its metric and its data all had problems, and its headline score did not measure what it claimed to.

1 Introduction

Face analysis systems such as recognition, landmarking and expression analysis degrade when part of the face is hidden. A segmentation mask of the occluded region lets a downstream model ignore or inpaint those pixels. Occluders are also semantic categories whose names carry strong priors about shape and position: "glasses" sit across the eyes and "face mask" covers the nose and mouth. Vision–language models such as CLIP already connect these words to image content, so we ask how far a small head on top of CLIP can go, given the name of the occluder as a prompt.

Contributions.

- A text-conditioned segmenter that turns CLIP's CLS-to-patch attention into a spatial prior and refines it with *Cross-Modal Attention Fusion* (CMAF) blocks driven by the prompt embedding (Section 4).
- A synthetic occlusion dataset with *exact* pixel labels, a held-out appearance split, and "negative" prompts that name an absent object (Section 3).
- An evaluation protocol that cannot be gamed by predicting nothing, or by ignoring the prompt (Section 5).
- Ablations, per-category and prompt-robustness analyses, and a failure analysis (Section 6).
- An account of what was wrong with v1 and how each problem was found (Section 7).



Figure 1: Generated training data. Odd rows: composited faces (CelebA-HQ) with glasses, face masks or both. Even rows: the exact labels (red: glasses, green: face mask), taken from the compositing step.

2 Related Work

Transformer-based segmentation. Fully convolutional networks [6] and encoder–decoders such as U-Net [5] established dense prediction from CNN features. Vision Transformers (ViT) [2] were extended to segmentation by SETR [7] and Segmenter [12] with dedicated decoders. We also use ViT features, but take the CLS-token attention map as a coarse cue instead of building a full decoder.

Attention maps as localisation cues. DINO [3] showed that self-supervised ViT attention roughly segments foreground objects, Chefer et al. [8] derived relevance maps for transformers, and Grad-CAM [9] localises with gradients. We use the last block's CLS-to-patch attention as a learnable prior: unlike these post-hoc methods, the blocks producing it are fine-tuned.

Vision–language segmentation. CLIPSeg [4] adds a text-conditioned decoder to CLIP; GroupViT [13] and OVSeg [19] target open-vocabulary segmentation and Grounded-SAM [14] chains a grounding model with SAM. Visus is much smaller and specialised: one CLIP backbone, a sub-million-parameter head, and a single binary mask per prompt.

Face occlusion. Occlusion-robust recognition has been approached with mask learning [11] and masked-face datasets [10]; MaskTheFace [17] synthesises masked faces. We reuse MaskTheFace for compositing and keep its alpha channel as the label.

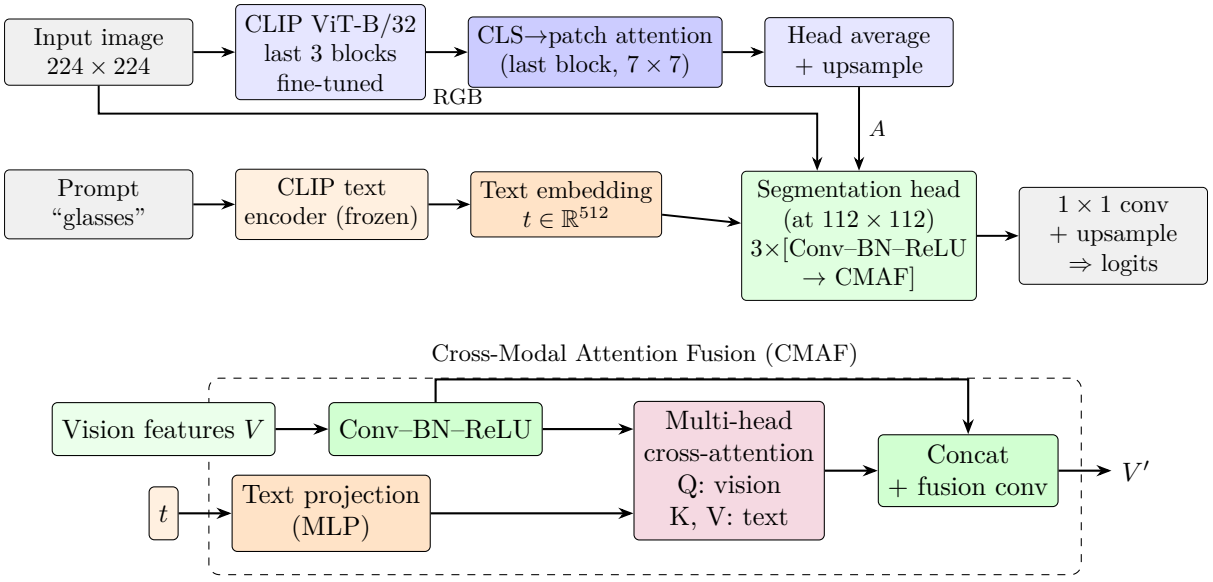


Figure 2: Top: Visus v2. The CLIP attention map and the RGB image enter a segmentation head that is conditioned on the prompt's CLIP text embedding. Bottom: one CMAF block.

3 Dataset

There is no public dataset with pixel masks for glasses and face masks, so we generate one from the 30,000 faces of CelebA-HQ [16] (Figure 1). For each face, dlib's 68 landmarks place the occluders:

- **Face masks.** MaskTheFace [17] warps one of its templates (surgical in three colours, N95, KN95, cloth) onto the face, with a texture or a random tint. We keep the alpha channel of the warped template, so the label is exactly the set of covered pixels.
- **Glasses.** Drawn procedurally with 4× supersampling: round, rectangular, wayfarer, aviator and cat-eye frames, with dark, mirrored, tinted or clear lenses, scaled and rotated to the eye landmarks. The label is the drawn footprint (lenses and frame).

40% of faces get glasses, 40% a face mask and 20% both; where they overlap, the glasses label is removed from the mask label. 332 faces with no detected face are dropped, leaving 29,668 images (17,884 with glasses, 17,656 with a mask, 5,872 with both). The median label covers 6.8% of the image for glasses and 12.4% for masks. Images are split 25,215 / 1,484 / 2,969 (train / val / test). **Five of MaskTheFace's textures are held out**: they never appear in training or validation, and every textured mask in the test split uses one of them.

Evaluation pairs. At test time every image is queried with *both* prompts, “glasses” and “face mask”. A pair whose object is present (3,542 pairs) is scored by IoU; a pair whose object is absent (2,388 pairs, e.g. “glasses” on a mask-only face) should produce an empty mask.

4 Method

Figure 2 gives an overview.

Visual prior. A forward hook on the last block of CLIP's ViT-B/32 image encoder recomputes its attention from the block's LayerNorm-ed input (exactly as CLIP computes it), and keeps the CLS token's attention over the 7×7 patch grid, $a \in \mathbb{R}^{B \times H \times 49}$. Averaging over the H heads and

upsampling gives the prior

$$A = \text{Upsample}\left(\tfrac{1}{H}\textstyle\sum_h a^{(h)}\right) \in \mathbb{R}^{B\times 1\times 224\times 224}. \tag{1}$$

The computation is differentiable, and the **last three vision blocks are fine-tuned**, so the backbone learns to aim its attention at the occluder; the other nine blocks stay frozen.

Text conditioning. The prompt is encoded by CLIP's **frozen** text encoder into an L2-normalised 512-d embedding t. Keeping it frozen preserves CLIP's knowledge of related phrasings (Table 5).

Segmentation head with CMAF. The head receives $[A; \text{RGB}]$ (4 channels) at half resolution (112×112, which cuts activation memory by $4\times$). Each of three stages is a Conv–BN–ReLU followed by a CMAF block: vision features are flattened into queries, the projected text embedding provides key and value of a multi-head cross-attention, and the result is concatenated with the vision features and fused by convolution,

$$V' = \text{Fuse}\left(V,\ \text{CrossAttn}(Q{=}V,\ K{=}V{=}\,\text{MLP}(t))\right). \tag{2}$$

A 1×1 convolution produces logits, upsampled to 224×224; $\sigma(\cdot)$ is the probability that a pixel is covered by the named object. In total 22.0 M parameters train: 21.3 M in the three CLIP blocks and about 0.7 M in the head.

Loss. BCE + Dice on the logits. Dice handles the class imbalance (an occluder covers 7–12% of the image); BCE gives a well-defined gradient on empty targets, where Dice alone is degenerate.

Negative prompts. During training, 25% of samples are queried with an object that is *not* in the image (when such an object exists), with an all-zero target. Without this, the model can learn to segment "whatever occluder is there" and ignore the prompt. Training prompts are drawn from seven phrasings per class (e.g. "sunglasses", "eyewear", "surgical mask").

5 Evaluation protocol

- **IoU** (occlusion IoU): mean per-pair IoU between the predicted and true *occluded* region over pairs where the object is present, reported for glasses, face masks, and their mean. An empty prediction scores 0.
- **Clean rate**: over pairs where the prompted object is absent, the share of predictions that are empty (under 0.5% of the image).
- **Score** $= \sqrt{\text{IoU} \times \text{clean rate}}$. It is 0 for a model that never predicts anything *and* for one that ignores the prompt. The score selects both the checkpoint and the binarisation threshold, on the validation split only.

Every table also reports an *empty-mask baseline* that predicts nothing. We added it after finding that v1's metric gave such a baseline a high score (Section 7).

6 Experiments

6.1 Setup

Table 1 lists the training configuration. Training ran on one NVIDIA T4 (Kaggle) and took 113 minutes. Before training, an automated check verified that gradients reach the three tuned CLIP blocks but not the frozen ones, and that a 6-image batch can be overfit (loss $1.48 \to 0.011$).

Table 1: Training configuration of Visus v2.

Backbone	CLIP ViT-B/32, last 3 blocks tuned	Epochs	20
Input size	224×224	Batch size	64
Optimiser	AdamW, weight decay 0.01	Schedule	1-epoch warmup + cosine
LR (head / CLIP blocks)	3×10^{-4} / 1×10^{-5}	Gradient clip	1.0
Loss	BCE + Dice	Precision	fp16 autocast
Augmentation	crop, flip, $\pm 10°$, colour jitter	Negative prompts	25%
Selection	best val score (epoch 18)	Threshold	0.6 (chosen on val)

6.2 Main results

Table 2: Held-out synthetic test split: 2,969 faces, 3,542 present and 2,388 absent pairs.

Model	IoU glasses	IoU mask	IoU	Clean rate	Score
Empty-mask baseline	0.000	0.000	0.000	1.000	0.000
v1 (`visus_epoch50.pt`)	0.357	0.467	0.412	0.001	0.019
Visus v2	**0.952**	**0.979**	**0.965**	**0.997**	**0.981**

Table 2 shows that v2 segments both occluders accurately and almost always stays silent when the prompted object is absent. v1, evaluated with the same protocol, recovers some occlusion but essentially never returns an empty mask: it predicts "an occluder" regardless of the prompt. Figure 3 shows that training is smooth: validation loss tracks training loss and the score plateaus after about 15 epochs.

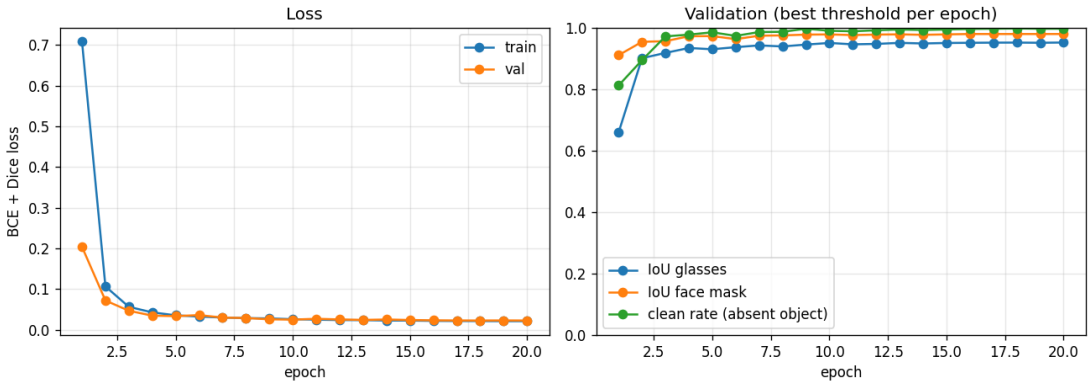


Figure 3: Training and validation loss (left); validation IoU and clean rate at the best threshold of each epoch (right).

6.3 Ablations

Table 3 removes one component at a time. To keep the cost manageable, all four variants are trained the same short way (6 epochs on 10,000 training images, otherwise identical settings), and each threshold is chosen on validation. They are then scored on the full test split. The full model is therefore weaker here than the released one, but the variants are comparable to each other.

- **Fine-tuning CLIP** is worth 8 points of score. With the backbone frozen, both IoU and the clean rate drop: the attention of a frozen CLIP is less focused on the occluder.
- **The RGB image** matters most for glasses (0.739 → 0.893 IoU). A 7×7 prior cannot resolve thin frames; the image provides the edges. The clean rate barely moves, because



Figure 4: The prompt decides what is segmented. Test faces wearing both occluders, queried with "glasses" (red) and "face mask" (blue). Masks here use textures held out of training.

Table 3: Ablations (short schedule, full test split).

Variant	IoU glasses	IoU mask	IoU	Clean rate	Score
Visus v2	**0.893**	**0.947**	**0.920**	**0.961**	**0.940**
CLIP frozen (no fine-tuning)	0.811	0.866	0.838	0.879	0.858
Head without the RGB image	0.739	0.850	0.794	0.946	0.867
v1 text features ($\mathbb{R}^3$)	0.468	0.483	0.476	0.004	0.042

deciding *whether* the object is there is the job of the attention prior and the text.

- **The text representation** is decisive. With v1's three scalars the clean rate collapses to 0.004: the head cannot tell the prompts apart, so it segments whichever occluder it sees. This reproduces v1's behaviour (Table 2).

6.4 Where does it fail?

Performance is uniform across styles (Table 4). Unseen mask textures cost only 0.004 IoU, so the model has not memorised mask appearance. The hardest glasses are those with clear lenses, where the frame is the only visible evidence. Figure 5 shows random test pairs and Figure 6 the worst ones. Most of the worst cases are thin or clear frames over a busy background, or faces where the original CelebA-HQ photo *already* contained real glasses or an eye-covering accessory. Those real objects are not labelled, so when the model segments them (the last two columns of Figure 6, prompt "glasses" with no synthetic glasses), it is counted as a false positive although it is arguably correct. Part of the remaining error is therefore label noise in the base photos.

Table 4: Test IoU by category (present pairs). Every textured mask in the test split uses one of the five textures held out of training.

Glasses	IoU	Face mask	IoU
round / wayfarer / aviator	0.956 / 0.955 / 0.951	surgical (3 colours)	0.977–0.981
cat-eye / rectangular	0.949 / 0.946	N95 / KN95 / cloth	0.978 / 0.981 / 0.978
lens: dark / mirror / tint	0.954 / 0.952 / 0.953	solid colour (seen)	0.980
lens: clear	0.941	**held-out texture**	**0.976**
alone / with a mask	0.954 / 0.948	alone / with glasses	0.980 / 0.976



Figure 5: Random test pairs: prompt, CLIP attention, prediction (red) and ground truth (green). The last two columns ask for an object that is not present; both predictions are empty.

6.5 Prompt robustness

Table 5 re-runs 1,000 present test pairs per class with other phrasings, including some never used in training. IoU is unchanged to the third decimal: the frozen CLIP text encoder maps synonyms close together, and the head responds to the meaning rather than the exact words.

The same table shows the limit of this. Prompts that name *neither* class ("hat", "beard", "a cat") should give an empty mask, but they do so only 0–39% of the time. The model has learned a two-way choice, and it maps any prompt to the nearer of its two classes. Visus is therefore a prompt-conditioned two-class segmenter, not an open-vocabulary one. Adding unrelated prompts to the negative examples during training is the obvious fix.

Table 5: Prompt robustness on 1,000 present pairs per class (* = phrasing never seen in training), and the empty rate for prompts that name neither class.

Glasses prompt	IoU	Mask prompt	IoU	Unrelated prompt	Empty
glasses	0.952	face mask	0.979	hat	0.30
sunglasses / eyewear	0.952 / 0.952	mask / surgical mask	0.979 / 0.979	beard	0.39
reading glasses*	0.951	N95 respirator*	0.979	a cat	0.16
aviator sunglasses*	0.952	covid mask*	0.979	a red car	0.00
eyeglass frames*	0.952	face covering*	0.979		

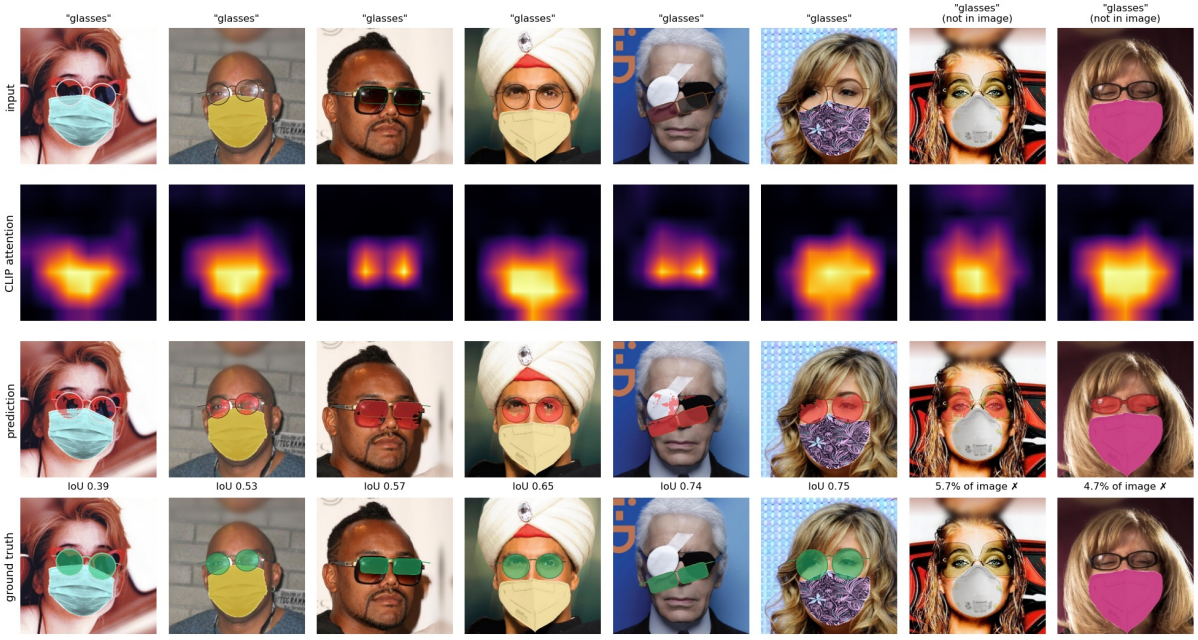


Figure 6: The six lowest-IoU present pairs and the two absent pairs with the largest predicted area. In the last two columns the base photo already shows real glasses, which carry no label.

6.6 v1 test sets (LFW and CelebA)

The original project's test subsets, 300 LFW [15] and 363 CelebA images with synthetic occlusions, have ground truth drawn as coarse polygons from MediaPipe [18] face landmarks rather than from the pasted occluders. The labels are therefore imprecise, and about 10% of the LFW mask labels are empty. On these sets v2 scores occlusion IoU 0.500 (LFW) and 0.485 (CelebA), against 0.417 and 0.515 for v1, and 0.053 / 0.020 for the empty baseline. These numbers mostly measure agreement with the polygon shape rather than segmentation quality. Figure 7 shows that v2's masks follow the actual glasses frames and masks on these images, while the reference polygons do not.

7 From v1 to v2: what was wrong

The April 2026 version of this report claimed mIoU 0.9441 (CelebA) and 0.9405 (LFW). Re-examining the code showed that these numbers did not measure occlusion segmentation, and that the model had not learned what we believed it had. Table 6 lists the problems and their fixes.

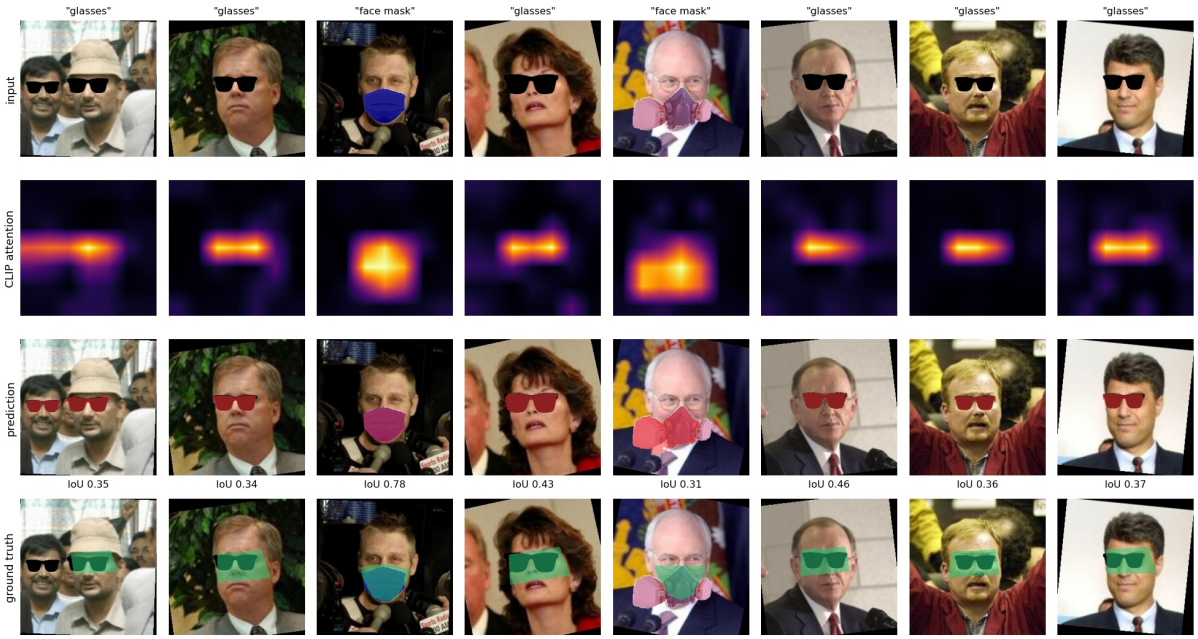


Figure 7: v2 on the v1 LFW test images. Row 3: prediction (red); row 4: v1 ground-truth polygons (green).

Table 6: Problems found in v1 and how v2 addresses them.

v1	v2
The model predicted the *visible face*, and IoU was computed on that background region. It covers most of the image, so a model that predicts no occlusion at all already scores highly.	Predict the occluded region; report occlusion IoU next to an empty-mask baseline.
The "fine-tuned" CLIP block received no gradient: the attention extractor ran under `no_grad` and detached its output. Only the head trained.	Differentiable extractor; a pre-training check asserts that gradients reach the tuned blocks.
Text features were three scalars (CLS attention per layer), carrying almost no information about the prompt.	512-d CLIP text embedding; negative prompts teach the model to respond to it (clean rate 0.001 $\rightarrow$ 0.997).
Training labels came from a Kaggle CelebA set whose masks turned out to be random face parts (eyes, lips, hair, skin), not glasses or masks. Test labels were landmark polygons.	Exact labels from our own compositing; a held-out texture split; inspection of the data before use.
The checkpoint was the last epoch, with a 0.9 threshold picked without validation; the loss curve spiked (Dice 0.0468 at epoch 10, 0.165 at epoch 18).	Warmup + cosine schedule, gradient clipping; checkpoint and threshold chosen on validation.
An ablation figure in the April report was not backed by experiments.	Removed. Every number in this version comes from a script in the repository, and the raw outputs are in `results/`.

8 Limitations and future work

- **Synthetic only.** All training and the main test use composited occluders. Real photographs (reflections, hands holding masks, partially worn masks, transparent frames) are not yet measured; a small hand-labelled real test set is the most important next step.
- **Two classes, closed vocabulary.** Only glasses and face masks are trained, and prompts

naming anything else are mapped to one of them (Table 5). Training with unrelated negative prompts, and with data for hats, scarves or hands, would make the prompt mechanism genuinely open.

- **Unlabelled real accessories.** Some base photos already contain real glasses, which carry no label; this adds noise to both training and the test numbers.
- **Resolution.** The prior is a 7×7 grid and the head runs at 112×112, so thin frames are where most errors are. A ViT-B/16 backbone or a skip connection from earlier blocks would help.
- **One seed.** Results are from single runs; we have not measured run-to-run variance.

9 Conclusion

Visus shows that a frozen CLIP text encoder, the attention of a few fine-tuned CLIP vision blocks and a small cross-modal head are enough to segment facial occlusions by name. It reaches 0.965 occlusion IoU on held-out synthetic faces, and returns an empty mask when the prompted occluder is absent. Much of the project's value came from making the evaluation honest: exact labels, a baseline that predicts nothing, and selection on validation. Those changes turned a reported 0.94 that measured the wrong thing into a 0.965 that measures the right one.

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